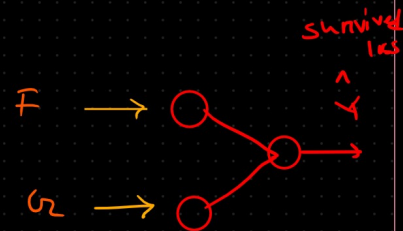


Agenda:

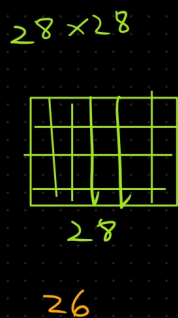
- ① RNN - Recurrent Neural Networks
- ② What is Sequential Data Learning?
- ③ Some Drawbacks in ANN and CNN
- ④ Types of RNN Networks
- ⑤ Practical Demo of RNN using Keras (Sentiment Analysis)

ANN → Tabular Data

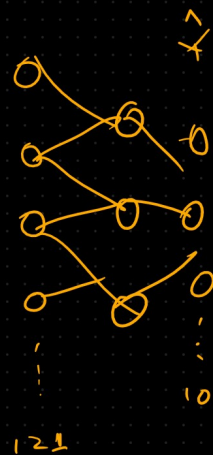
X		Y
Q2	Fare	Survived
F	20K	1
M	5K	0
-	-	-
-	-	-



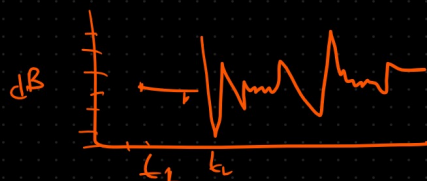
CNN → Image 28



11x11
⇒ 121



- ① My name is Buppy
 t_1 t_2 t_3 t_4
- ② I love AI



→ Text

→ Time Series

→ Speech

→ DNA seq

* Why should use RNN?

5	Hi my name is Bappy	0
3	I love AI	1
4	Bappy is my name	0

Total: 12

Hi $\Rightarrow [1, 0, 0, 0, \dots, 12]$

My $\Rightarrow [0, 1, 0, 0, \dots, 12]$

(2)

$[1, 0, 0, \dots, 12]$

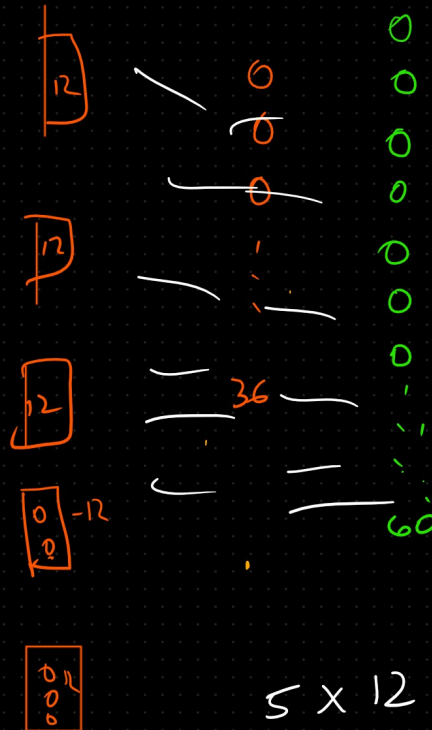
$[0, 1, 0, \dots, 12]$

$[0, 0, 1, \dots, 12]$

$12 \times 3 \Rightarrow 36$
 $12 \times 4 \Rightarrow 48$

Hand eated solution:

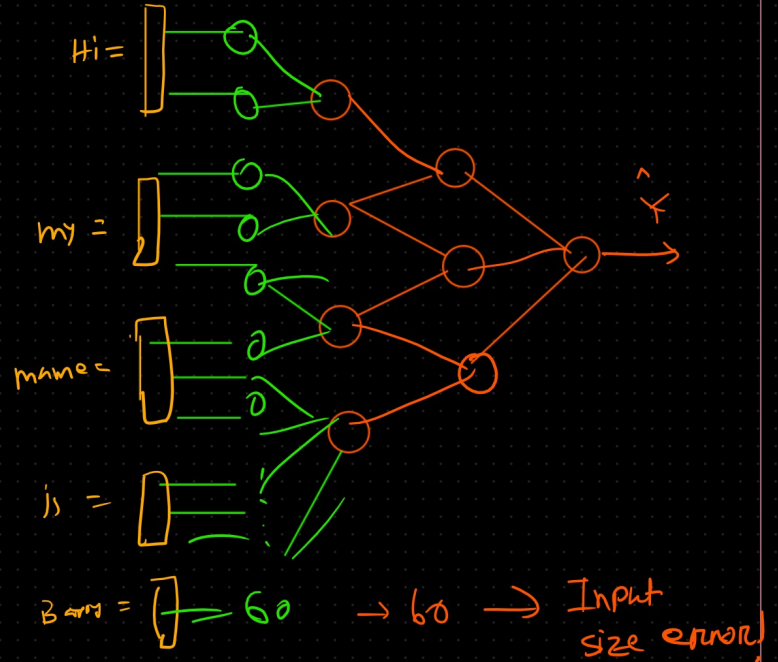
28×28
 32×2



$\rightarrow 60 \rightarrow$ Input size error

$5 \times 12 \Rightarrow 60$

$12 \times 5 = 60$



* Note all problems:

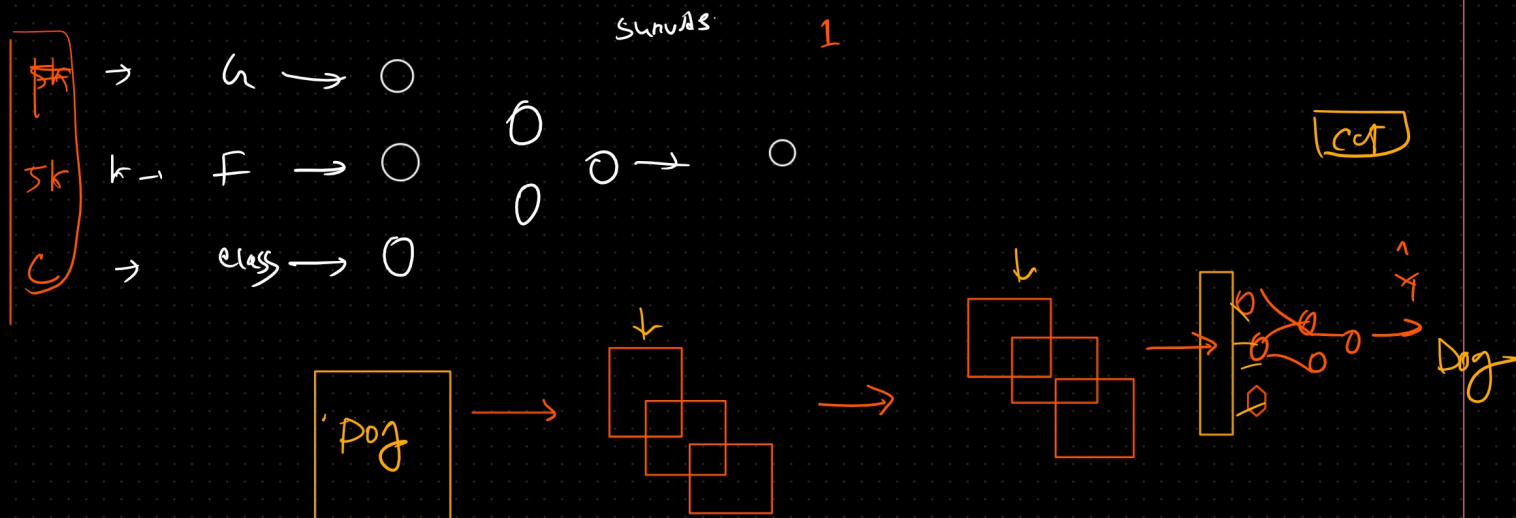
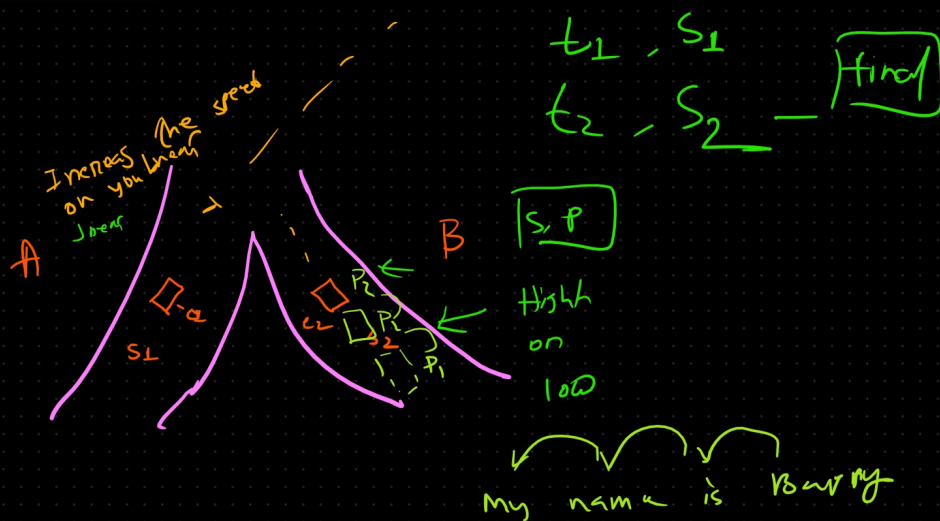
- ① Text input $\rightarrow$ varying size
- ② zero padding $\rightarrow$ sparse matrix $\rightarrow$ unnecessary computation
- ③ losing sequential info (context mech)

RNN - Recurrent Neural Network

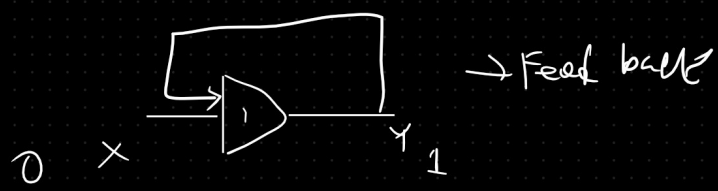
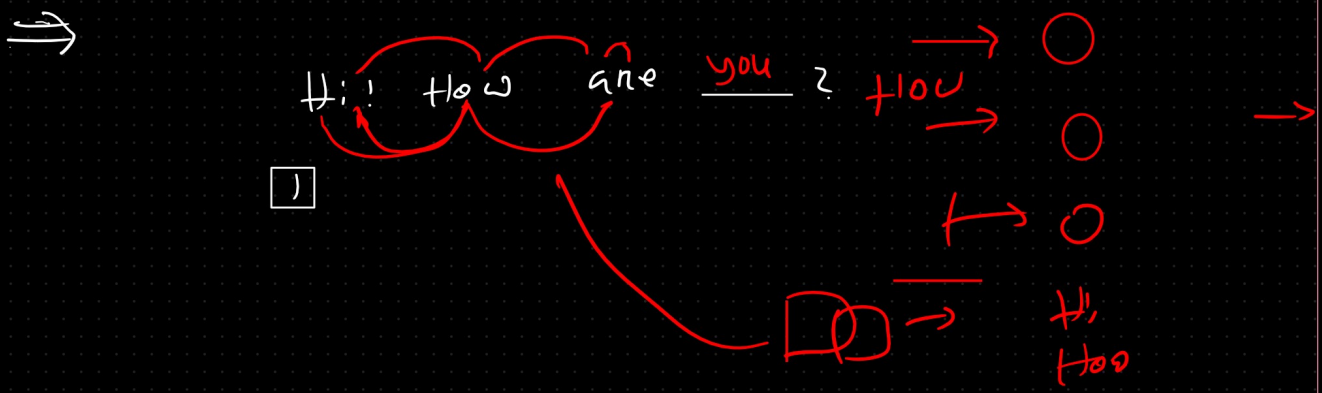
$\rightarrow$ 1980

Demo:

Self driven $\rightarrow$
RL $\rightarrow$



Hi How are you?



RNN → Recurrent Neural Network

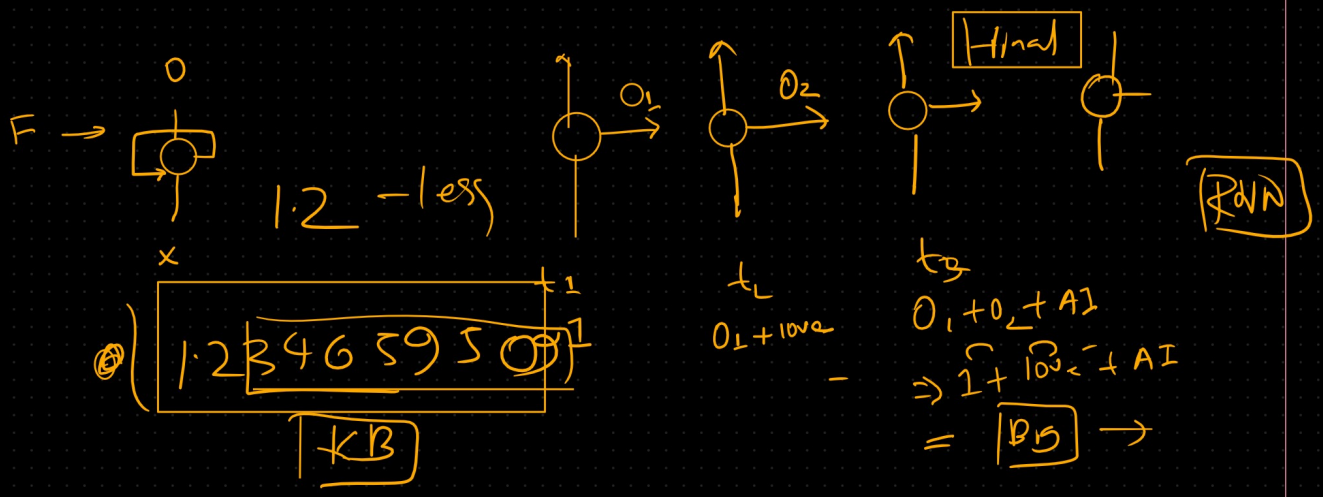
Recursion

DSA



[I love AI]

[My man is here]



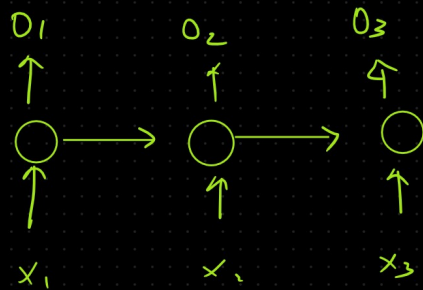
1.20345657 → Hight KB

1.2 → less → KB



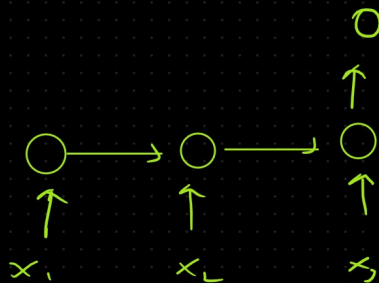
Types of RNN/sequential Architecture:

Many to many: (sec to sec)



⇒ Translator

Many to one (sec to vec)

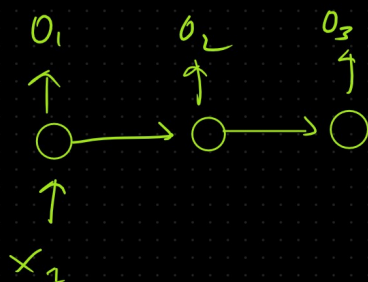


⇒

Text classification

○
○
○

One to many (vec to sec):



⇒

Text generation

One to one (vec to vec):

